

Metropolitan University

Department of Computer Science and Engineering
Summer Term Final Examination – 2025

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Program: B.Sc. in CSI; **Batch:** 63 (All section)
Course: GED 202: History of Emergence of Bangladesh

Time: 2 Hours

Marks: 40

Answer any **four** sets of questions. All parts of each question must be answered sequentially.
Figures in the right margin indicate full marks.

- ✓ 1. a) Explain the activities and influence of different European powers in Bengal during the early modern period. 5
b) What were the political and economic reasons behind the conflict between the British East India Company and Nawab Siraj-ud-Daulah that led to the Battle of Plassey? 5
2. a) Outline the core issues that led to the Revolt of 1857. 5
b) Explain the primary reasons for the failure of the Revolt of 1857. 5
- 3. a) Discuss the main features of Lahore Resolution. Do you believe that the Lahore Resolution contained the seed for Bangladesh's independence? 5
b) Explain the main reason why Bengal was partitioned in 1905. 5
- 4. a) Describe the key factors that led the Language Movement of 1952 in East Pakistan. 6
b) What is the significance of the Language Movement of 1952 in shaping the identity and future of East Pakistan? 4
- 5. a) State the causes that led to the Mass Upsurge of 1969 in East Pakistan. 5
b) Illustrate the results of the 1970 Pakistani general election and explain how they influenced the Liberation War of 1971. 5
6. Write Short notes on: (Any Two) 10
a) United Front election of 1954
b) Exile Government of Bangladesh
c) Operation Searchlight
d) Surrender at Dhaka

Metropolitan University

Department of Computer Science and Engineering

Summer Term Final Examination – 2025

Program: B.Sc. in CSE; Batch – 63 (All Sections)

Course: MAT 112: Differential and Integral Calculus

Time: 2 Hours

Marks: 40

Answer any four questions. All parts of each question must be answered sequentially. Figures in the right margin indicate full marks.

1. (a) Define with example: i) Injective function ii) Even and odd functions 2
(b) Find domain and range: i) $y = \frac{2}{3x-1}$ ii) $y = \sqrt{x^2 + 25}$ iii) $y = \frac{1}{x^2 - 16}$ 5
(c) Sketch the following function using graph shifting technique: 3
 $y = x^2 + 16x + 64$
2. (a) Consider the following functions and determine where the limit doesn't exist: 4
i) $f(x) = \frac{x-1}{|x-1|}$ ii) $f(x) = \begin{cases} 0 & \text{if } x < 3 \\ 1 & \text{if } x \geq 3 \end{cases}$
(b) Prove that $f(x) = \begin{cases} \frac{x^2-4}{x-2}; & x < 2 \\ \frac{3x+2}{x}; & x > 2 \\ 6; & x = 2 \end{cases}$ is discontinuous at $x=2$. 3
(c) Test the differentiability of $f(x) = \begin{cases} 5x-1; & x < 1 \\ 3x+x^2; & x \geq 1 \end{cases}$ at $x=1$. 3
3. (a) Examine $f(x) = 2x^3 - 21x^2 + 36x - 20$ for maximum and minimum values. 4
(b) Find the equations of the tangent and normal to the curve $y = x^2 + 6x + 9$ at the point $(-2, 1)$ 3
(c) State and Verify Rolle's theorem for the function $f(x) = \frac{x^2-6x}{x+2}$ in the interval $[0, 6]$. 3
4. Evaluate the integral (Any two) 10
(a) $\int \frac{1}{4x^2+4x+5} dx$ (b) $\int \frac{8x^2-19x+1}{(2x+1)(x-2)^2} dx$ (c) $\int \frac{x^3}{(x-1)(x-2)(x-3)} dx$
5. (a) Evaluate the definite integral: $\int_0^2 x^3 e^{x^2} dx$ 5
(b) Find the volume of the solid obtained by rotating the region bounded by $y = x^3$, $y = 8$, $x = 0$ about y axis. 5
6. (a) Evaluate the differential coefficients ($\frac{dy}{dx}$): 5
i) $y = \frac{(x^2+1)^6}{(x^2+2)^3}$ ii) $(\cos x)^y = (\sin x)^x$
(b) If $y = e^{m \sin^{-1} x}$ then prove that: 5
 $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (m^2+n^2)y_n = 0$

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Department of Computer Science and Engineering

Summer Term Final Examination – 2025

Program: B.Sc. in CSE; Batch: 63rd

Course: Eng 114: English Language I

Time: 2 Hours

Marks: 40

Read the passage and answer questions 1 and 2:

The future of AGI is an exciting and rapidly evolving field. While the development of AGI remains a challenge, it has the potential to revolutionize many aspects of our lives, from healthcare to transportation to education. One potential avenue for advancing AGI is through the creation of more powerful and sophisticated AGI foundation models. Recent breakthroughs in natural language processing, computer vision, knowledge graph, and reinforcement learning have led to the development of increasingly advanced AGI models such as ChatGPT and GPT-4. These models have shown impressive capabilities in various applications. Further advances in AGI foundation model research, as well as improvements in hardware and computational algorithms, are very likely to accelerate the development of AGI. Another approach to developing AGI is through the integration of different AI systems and technologies across multiple domains, including adding human in the loop through reinforcement learning from expert feedback. This integration could also help overcome the limitations of current AI systems, which are often specialized in specific domains and lack the flexibility to transfer knowledge across domains. The development of AGI also requires the development of novel approaches for machine learning, such as more efficient instruct methods, in-context learning algorithms, and reasoning paradigm, particularly by learning from the human brain via brain-inspired AI. These approaches aim to enable machines to learn from unstructured data without the need of labeling them and rapidly generalize from a few examples, which is crucial for enabling machines to learn and adapt to new tasks and environments. Finally, ethical and societal implications of AGI development must be considered, including issues related to bias, privacy, and security. As AGI becomes more powerful and pervasive, it is essential to ensure that it is developed and used in a responsible and ethical manner that benefits society as a whole and aligns well with human value. Overall, while the development of AGI remains a challenge, it has the potential to revolutionize many aspects of our lives and bring significant benefits to society and humanity. Ongoing research and development in AGI will continue to drive progress towards the ultimate goal of creating truly intelligent machines.

(Source: <https://doi.org/10.1016/j.metrad.2023.100005>)

1. Fill the gaps with ONE word only:

1×5=5

- A. Latest innovations in reinforcement _____ have contributed to the progress in AGI development.
- B. The amalgamation of multiple AI systems can help conquer the _____ of current domain-specific models.
- C. Brain-inspired AI is one approach that mimics _____ learning to improve AGI adaptability.
- D. Machines should be able to learn from _____ data without needing explicit labeling.
- E. AGI systems must be designed in a way that respects morality like prejudice, secrecy, and _____.

2. Answer the following questions:

1×5=5

- A. What are some of the recent breakthroughs that have led to the development of advanced AGI models?
- B. According to the text, what is the potential benefit of integrating different AI systems and technologies?
- C. What are the novel approaches for machine learning that are needed to enable machines to learn and adapt to new tasks?
- D. What are some of the ethical and societal implications of AGI development that must be considered?
- E. How can further advances in AGI foundation models, hardware, and computational algorithms accelerate the development of AGI?

3. Fill in the blanks with the correct verb form (a-e) and appropriate prepositions (f-j). 0.5×10=5

The rise of artificial intelligence in modern industries (a) _____ (bring) significant changes to the way businesses operate. While some sectors (b) _____ (experience) rapid growth due to automation, others (c) _____ (struggle) to adapt to the evolving technological landscape. The lack of skilled professionals in this field often (d) _____ (cause) delays in project execution, and not every organization (e) _____ (be) prepared for this shift.

Many experts emphasize the importance of investing (f) _____ education and training programs. Without proper guidance, employees may struggle (g) _____ new tools and systems.

Governments around the world are working (h) _____ policies that can support both innovation and employment. In such a dynamic environment, success depends largely (i) _____ how quickly individuals can learn and adjust. Collaboration (j) _____ different sectors is now more essential than ever.

4. Complete the following sentences with appropriate conditional forms. 0.5×10=5

- A. If we had a bigger house, we _____.
- B. She would travel more if she _____.
- C. If it rains tomorrow, we _____.
- D. If I had known about the party, I _____.
- E. If you had listened to me, you _____.
- F. If we don't hurry, we _____.
- G. If they had left earlier, they _____.
- H. If we talked about the matter earlier, _____.
- I. Rafi could not have texted me if _____.
- J. If I was there at the station, _____.

5. Make five sentences using any five items from the given list. 1×5=5

Be at odds with, The best of both worlds, Spread the word, drop back, pass out, pay for something, get around.

6. Suppose you are Robin/Ritika, a CSE graduate from MU, found a circular for the post of senior programmer in Amazon via LinkedIn on 20th July, 2025. Now write a cover letter for this job. 1×5=5

Or

Suppose you have bought a VR set from a local Samsung outlet. But you are facing some sound issues. Now write a complaint letter to the authority.

7. Write any ONE of the following:

1×10=10

A) Role of Networking in career

B) Role of Youth in society

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Department of Computer Science and Engineering

Summer Term Final Examination – 2025

Program: B.Sc. in CSE; Batch: CSE 63

Course: CSE 125: Discrete Mathematics

Time: 2 Hours

Marks: 40

Answer any **four** sets of questions. All parts of each question must be answered sequentially.

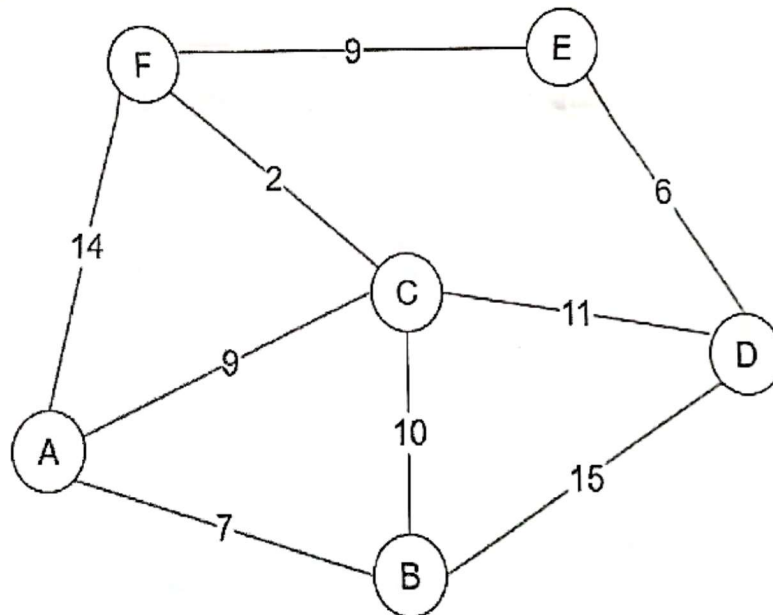
Figures in the right margin indicate full marks.

55	41	32	26	48	65	59	50	72	99	84	93	77	62	44	49
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Fig 1

- a) Perform **binary search** on the following **Fig 1** and find the index of value **93** (if the value exists in the array). Show all steps, and show start, end and mid at every step. 5

- b) Create a Binary search tree from **Fig 1**. 5



- a) Find the **adjacency matrix** and **adjacency list** representations for the given graph. 5

- b) Use **Dijkstra's algorithm** to find the shortest distance from vertex **A** to all other vertices. 5

- a) Factorise 2009 using Fermat's algorithm. 5

- b) Create pairs from the following number based on congruency(mod 13)? 5
39, -26, 65, 102, 51, 27, 14, -12, 78, 50

Abide

4. a) Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, $P = \{1, 2, 5, 9\}$, $Q = \{5, 6, 7, 10\}$.
 i) $P \cap Q$ ii) $Q - P$

2

- b) Determine whether these system specifications are consistent or not:
 Chandler does not do funny things and Joey wears colorful dresses. Monica does not like cleaning if and only if Chandler does funny things. Monica likes cleaning or Rachel likes rain. Rachel does not like rain. If Ross is happy then Rachel likes rain.

4

- c) Show that $p \leftrightarrow q$ and $(p \rightarrow q) \wedge (q \rightarrow p)$ are logically equivalent.

4

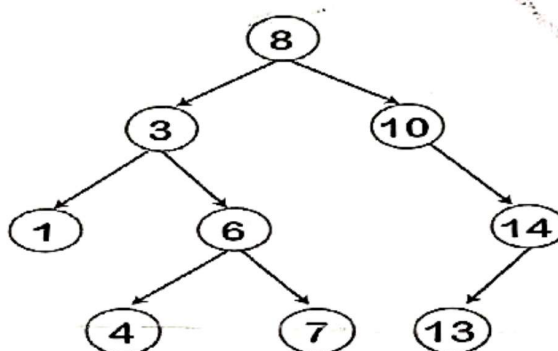
5. a) Decrypt the ciphertext message " WKLQN GLVFUHHWHOB VROYH
 HOHJDQWOB " using caesar cipher.

5

- b) Use the Euclidean algorithm to find $\gcd(78945, 45678)$.

5

6.



Graph 1

- a) What is the difference between a graph and a tree?

2

- b) From Graph 1, identify the following:

3

- Leaf nodes
- Root node
- Depth of the tree

- c) Determine the **preorder**, **inorder**, and **postorder** traversals of the Graph 1 and determine if any of the traversal results in a sorted sequence of numbers.

5

Metropolitan University

Department of Computer Science and Engineering

Summer Term Final Examination – 2025

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Program: B.Sc. (Engg.) in CSE; Batch: 63

Course: PHY 111: Physics 1

Time: 2 Hours

Marks: 40

(Answer any four questions. Please answer all parts of each question sequentially.)

1. a) What is a mechanical wave? **Discuss** the fundamental differences between a Longitudinal and a Transverse wave. 4
- b) **Calculate** the (i) wavelength, (ii) period, (iii) amplitude, and (iv) phase constant of the traveling wave equation given by, 4
- $$y(x, t) = (20.80 \text{ cm}) \sin(22.8\pi x - 66.43\pi t - 0.25\pi)$$
- c) **What** is the direction of the wave? **Write the equation** of another wave that travels in the opposite direction with a zero phase constant and half of the amplitude of the wave [(given in question (b))]. 2
2. a) **What** are the energies of a harmonic oscillator? **Draw the energies** with time on a graph and indicate the kinetic, potential, and total energies. 2+2
- b) **Find** the (i) velocity and (ii) maximum acceleration of the simple harmonic oscillator described by the equation, 4
- $$X = x_m \cos(kx - \omega t)$$
- c) **Draw the graphs** of velocity and acceleration with time that were calculated in question 2(b). 2
3. a) **Distinguish** between the three types of heat transfer methods. **What** are the latent heat of fusion and the latent heat of vaporization? 3+2
- b) **How** the anomalous expansion of water from 100°C to 0°C helps protect marine life during the winter, when the oceans at the north and south poles are covered with ice. 2
- c) 100g of ice is melted from 0°C to liquid water and vaporized at 100°C. **Find the total heat** absorbed during the melting and vaporization. (Latent heat of fusion, $l_f = 336 \text{ KJ/Kg}$, and latent heat of vaporization, $l_v = 226 \text{ KJ/Kg}$). 2+1
- Is the fusion of ice and the vaporization of water isothermal or adiabatic?
4. a) **Draw** the adiabatic and the isothermal curves on a PV diagram. **Why** is the adiabatic line steeper than the isothermal line? 2
- b) **Find** the work done in an adiabatic process when 5 moles of an ideal gas at a pressure of 100 Pascal and a temperature of 250°C is expanded 5 times its initial volume. Take $\gamma = 1.4$. 4
- c) **Show** that the change in the internal energy of a gas is proportional to the change in temperature. 3

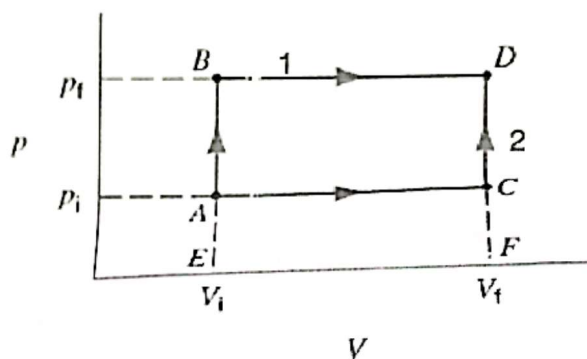
Please turn over.

- 5 a) What do you mean by the statement, "the specific heat capacity of a substance is 450 J/kg K"? Show that the molar heat capacity of a gas at constant volume is

$$c_v = \Delta E_{int} / n \Delta T$$

where, ΔE_{int} is the change in internal energy, n is the molar mass, and ΔT is the temperature change.

- b) One mole of gas expands from 1.0 m^3 to 5.0 m^3 while its pressure decreases from 15 to 5.0 Pa. Calculate the work done on the gas if the temperature remains unchanged at 100°C while the volume change takes place. Universal gas constant $R = 8.31 \text{ J/(mol-K)}$ 3
- c) Draw two separate graphs and show the work done by a gas for (i) path 1 (A-B-D) and (ii) path 2 (A-C-D). See the PV diagram given below. List the thermodynamic processes involved in the two paths.. 2+1



- 6 a) A Carnot Engine takes heat from the hot reservoir in an isothermal process and then undergoes an adiabatic expansion. Then the engine goes through another isothermal process and adiabatic compression. What will be the efficiency of the Carnot engine if the two isothermal processes (T_H and T_L) take place one after another 2
- b) Explain the working procedure of a Carnot Engine using an Entropy - Temperature diagram and calculate the efficiency of the engine. 2+2
- c) Calculate the (i) efficiency and (ii) work done per engine cycle of a Carnot engine that absorbs 1250 kJ of heat and exhausts 800 kJ in each cycle. 4